

Dynamic Risk Portfolio Rebalancer

Risk Migration Monitoring Platform · LendingClub MVP · Version 1.0

Morgan Stanley Goldman Sachs JPMorgan Chase UBS HSBC Citi Barclays

01 Executive Summary

Enterprise lending portfolios carry a structural blind spot: risk assessment is concentrated at origination and rarely updated continuously thereafter. Once a loan is approved, the borrower's risk profile is typically monitored only through lagging indicators — missed payments, delinquency flags — by which point loss mitigation is already expensive or impossible.

The **Dynamic Risk Portfolio Rebalancer (DRPR)** addresses this gap. It is not a credit scoring system. It is a *risk migration and drift monitoring platform* that continuously computes the divergence between a loan's original underwriting risk estimate and its current inferred risk state, using repayment progression signals derived entirely from the LendingClub portfolio dataset.

CORE PROBLEM	SOLUTION TYPE	DATA SOURCE	TARGET USERS
Post-approval risk blindness	Risk drift detection	LendingClub loans	Risk officers, CROs

The core equation: **Underwriting Grade + Repayment Progression + Vintage Risk = Dynamic Risk Assessment**. This is not transaction monitoring. It is longitudinal risk migration analysis.

02 Business Problem

Traditional banking credit infrastructure operates as a one-shot pipeline: an application is received, underwriting assigns a risk grade, and the loan is approved or denied. Beyond approval, monitoring is reactive — triggered by delinquency events rather than by early detection of deteriorating borrower health.

The Gap

- A borrower graded B at origination (historical default rate: 13.39%) may have meaningfully deteriorated to what would be a D-equivalent risk level (30.38%) within 18 months.

- Without continuous reassessment, the portfolio's true risk-weighted exposure is systematically understated.
- Risk officers cannot distinguish between a loan that is performing according to underwriting expectations versus one that is drifting toward default silently.
- Provisioning, capital allocation, and intervention triggers all rely on stale origination-era risk grades.

What DRPR Does Not Do

DRPR does not replace underwriting. It does not score new applications. It operates entirely post-origination on an existing book of loans, monitoring the divergence between the underwritten PD and the current estimated PD.

03 Solution Overview

DRPR is a four-layer analytical architecture built on top of the LendingClub loan dataset. Each layer contributes a distinct signal to the final dynamic risk assessment.

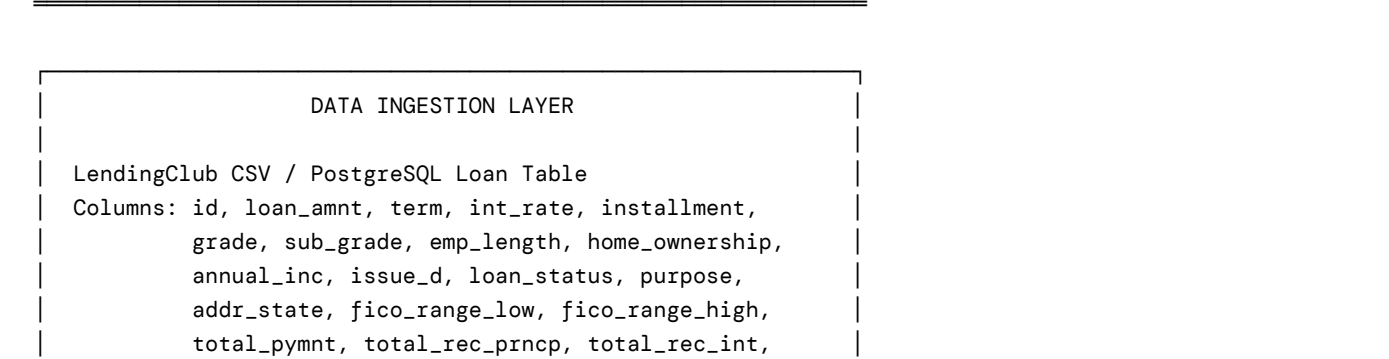
LAYER 1	LAYER 2	LAYER 3	LAYER 4
Underwriting Anchor grade, sub_grade → origination PD	Repayment Progression Payment ratios, months since last payment	Vintage Risk issue_d cohort macro adjustments	Latent State HMM Hidden borrower states: Healthy → Default

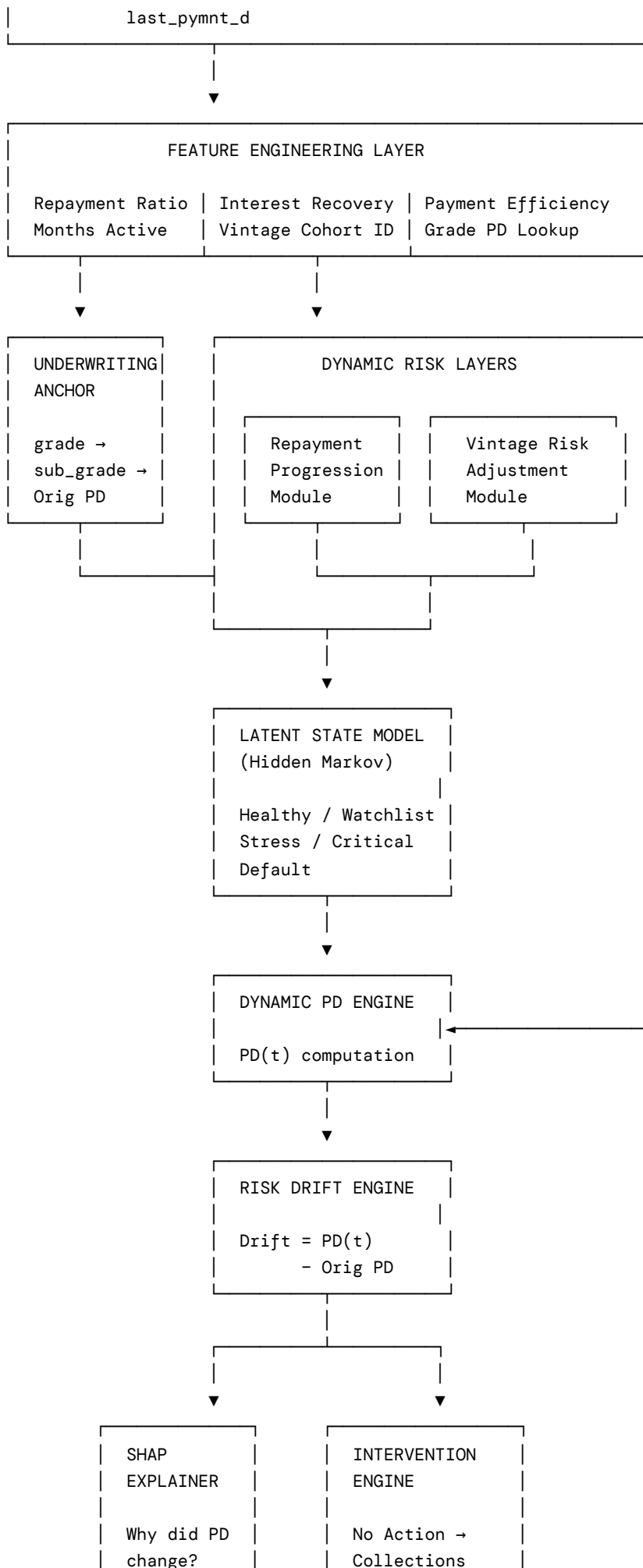
These four layers converge in the **Dynamic PD Engine**, which computes PD(t) — a time-varying probability of default that is then compared against the origination PD to produce the **Risk Drift Score**. That score drives the **Intervention Engine**, which recommends risk-tiered actions.

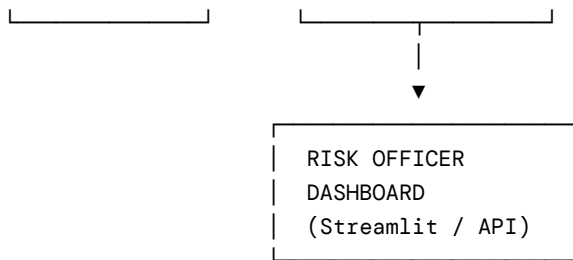
04 End-to-End Architecture

The end-to-end system flows from raw LendingClub data through progressive analytical enrichment layers, culminating in risk officer dashboards and automated intervention recommendations.

DRPR – END-TO-END ARCHITECTURE FLOW







05 Data Architecture

All MVP data flows originate exclusively from the LendingClub loan dataset. The following table maps each column to its functional role within DRPR.

COLUMN	ROLE IN DRPR	LAYER
grade, sub_grade	Origination PD anchor — primary risk calibration source	Underwriting
loan_amnt	Denominator for repayment ratio; portfolio exposure sizing	Repayment
total_rec_prncp	Numerator for repayment ratio and payment efficiency	Repayment
total_pymnt	Denominator for interest recovery and payment efficiency	Repayment
total_rec_int	Numerator for interest recovery ratio	Repayment
issue_d	Vintage cohort identification; months-active computation	Vintage
last_pymnt_d	Months since last payment; recency signal for latent state	Repayment / Latent
loan_status	Default ground truth (Charged Off = 1, Fully Paid = 0)	Target / Validation
int_rate	Risk-adjusted pricing signal; model feature	Underwriting
fico_range_low, fico_range_high	Origination credit quality; underwriting anchor feature	Underwriting
annual_inc, emp_length	Borrower capacity features for static PD model	Underwriting
term, installment	Scheduled repayment context; efficiency baseline	Repayment
purpose, addr_state	Segmentation features; geographic risk stratification	Segmentation
home_ownership	Collateral proxy; stability signal	Underwriting

No external data sources are used in the MVP. All features are derived entirely from columns present in the LendingClub dataset.

06 Underwriting Anchor Framework

The underwriting anchor is the origination-era risk estimate that DRPR uses as a baseline against which current risk is measured. LendingClub assigns each loan a grade (A–G) and a finer sub_grade (A1–G5), which represent the platform's assessment of borrower credit risk at origination.

Empirical historical default rates by grade are treated as the ground truth origination PD:

GRADE	ORIGINATION PD	RISK LEVEL	DEFAULT BAR
A	6.04%	Prime	
B	13.39%	Near-prime	
C	22.44%	Subprime	
D	30.38%	Subprime	
E	38.48%	High risk	
F	45.20%	High risk	
G	49.93%	High risk	

Sub-grades (e.g., B1 through B5) allow finer interpolation within each grade band. The sub_grade therefore provides a more precise origination PD anchor than grade alone. This 35-point sub_grade scale becomes the starting risk coordinate for every loan in the portfolio.

The underwriting anchor is **fixed at origination**. It does not change. It is the reference point against which all subsequent risk drift is measured. The anchor is only as good as the original underwriting — a critical limitation discussed in Section 18.

Origination PD Construction

For the MVP, origination PD is constructed as a lookup table mapping sub_grade → empirical default rate, interpolated linearly within each grade band. A logistic regression trained on origination-era features (grade, sub_grade, fico_range_low, fico_range_high, annual_inc, int_rate, dti, emp_length) serves as a secondary anchor to validate and refine these lookup values.

07 Repayment Progression Framework

Repayment progression is the primary dynamic signal available in the LendingClub dataset. As a loan ages, the cumulative repayment trajectory encodes meaningful information about the borrower's ability and willingness to service debt — the two core drivers of credit risk.

Four engineered metrics form the repayment signal vector:

Repayment Ratio

$\text{total_rec_prncp} / \text{loan_amnt}$

Measures what fraction of the original principal has been recovered. Low values relative to loan age indicate underpayment or payment shortfalls — a strong early-warning signal of stress.

Interest Recovery Ratio

$\text{total_rec_int} / \text{total_pymnt}$

Measures the proportion of total payments composed of interest rather than principal. Higher-than-expected interest share may indicate the borrower is paying minimum amounts only, reducing principal reduction rate and extending effective duration risk.

Payment Efficiency

$\text{total_rec_prncp} / \text{total_pymnt}$

The complement of interest recovery. High efficiency means principal is being retired aggressively, implying a borrower performing ahead of schedule. Low efficiency signals concern about debt service capacity.

Months Since Last Payment

$\text{months_between}(\text{last_pymnt_d}, \text{reference_date})$

Payment recency is a powerful leading indicator. A borrower who made a payment 1 month ago has materially different risk from one whose last payment was 5 months ago, even if cumulative amounts are similar. This metric bridges repayment history to the latent state model.

Months Active (loan age)

$\text{months_between}(\text{issue_d}, \text{reference_date})$

Normalizes all repayment metrics by loan age. A repayment ratio of 0.4 on a 12-month-old 36-month loan is on-track; the same ratio on a 30-month-old loan signals underpayment. All metrics must be interpreted in the context of expected amortization schedule.

Expected vs Actual Repayment

Each loan has a scheduled amortization profile computable from `loan_amnt`, `int_rate`, and `term`. The deviation between expected cumulative principal at month `t` and actual `total_rec_prncp` is the **repayment deviation signal** — arguably the most direct proxy for borrower financial health available without transaction-level data.

Not all loans of the same grade carry identical risk. A Grade B loan originated in 2014 — before the 2015–2016 LendingClub credit standard tightening — may have systematically different default characteristics than a Grade B loan from 2017, even if the borrower profiles appear similar. This is vintage risk: the macroeconomic and underwriting-standard environment embedded in a loan cohort at origination.

Vintage Identification

Vintages are defined using `issue_d`, grouped by year-quarter (e.g., 2014-Q1, 2015-Q3). Each vintage cohort accumulates a realized default rate as loans mature and resolve. This realized default rate becomes a vintage-specific adjustment factor applied to the origination PD.

Vintage Adjustment Mechanics

Vintage Default Rate

$$\text{vintage_PD} = (\text{charged_off_in_cohort}) / (\text{total_loans_in_cohort})$$

Computed per `issue_d` cohort. As a cohort matures, this rate converges toward the true lifetime default rate of that vintage. Earlier (more mature) cohorts provide more reliable estimates.

Vintage Adjustment Factor

$$\text{vintage_adj} = \text{vintage_PD} / \text{expected_PD_for_grade_mix}$$

Normalizes the vintage PD by the grade-mix-weighted expected PD. A value >1 indicates the vintage underperformed expectations (adverse macro environment); <1 indicates outperformance.

Macro conditions that embed into vintages include: interest rate environments, unemployment cycles, housing market cycles, and platform-specific underwriting policy shifts. The 2014 and 2015 vintages, for instance, contain loans originated under looser credit standards than 2016–2017, resulting in systematically higher realized default rates controlling for grade.

Vintage risk is a **portfolio-level correction**. Individual loan PDs are multiplied by the vintage adjustment factor to account for cohort-level macroeconomic effects that were not fully captured at origination. This prevents systematic mispricing of risk in high-risk vintage books.

09 Latent State Modeling Framework

Borrower financial condition does not change instantaneously — it evolves through a sequence of states. A borrower typically transitions from a healthy condition through a watchlist phase, then into stress, and potentially to critical or default over a period of months. These transitions are not directly observed in the dataset. They must be inferred from the observable repayment signals.

This is the foundational motivation for a **latent state model**: a layer of hidden variables that bridge observable payment metrics and future default risk.

Defined Latent States

Five discrete hidden states are defined. These states do *not* exist in the dataset — they are an abstraction layer constructed by the model:

Healthy	Repayment on or ahead of schedule. High payment efficiency. Recent payment activity. Low risk drift. No intervention required.	PD(t) $\approx$ Orig PD
Watchlist	Slight underpayment vs schedule. Payment efficiency declining. Monitoring flagged. Soft outreach recommended.	Drift 0–10%
Stress	Meaningful deviation from repayment schedule. Months since last payment increasing. Proactive intervention required.	Drift 10–25%
Critical	Severe underpayment. High payment gap. Collections escalation triggered. High PD(t).	Drift >25%
● Default	Charged Off event observed. Ground truth default. Feeds model retraining pipeline.	loan_status = CO

Hidden Markov Model (HMM) Architecture

A Hidden Markov Model with 5 hidden states is trained on the observable emission sequence: {repayment_ratio, payment_efficiency, months_since_last_payment, repayment_deviation}. The HMM learns:

- **Transition probabilities** P(state t+1 | state t): the probability of moving between states in a given period
- **Emission probabilities** P(observations | state): the distribution of observable signals given each hidden state
- **Initial state probabilities**: influenced by underwriting grade and sub_grade

The Viterbi algorithm is used to decode the most probable state sequence for each loan. Forward-backward inference provides the posterior state probability distribution at any point in time, which feeds directly into PD(t) computation.

HMM implementation uses **hmmlearn** (GaussianHMM for continuous emissions). The model is trained per-grade segment to account for the fact that healthy Gaussian state emissions differ materially between a Grade A and Grade E borrower.

10 Dynamic PD Engine

The Dynamic PD Engine synthesizes all upstream layers into a single time-varying probability of default, PD(t), computed at any observation point during the loan lifetime.

Origination PD

Origination PD is fixed at loan issuance. It is derived from the grade and sub_grade lookup table of empirical historical default rates, optionally refined by a static logistic regression trained on origination-era features. This is the anchor value — it does not change.

Current PD(t) Computation

PD(t) is computed as a weighted combination:

PD(t) — Dynamic Probability of Default

$$PD(t) = f(Orig_PD, Repayment_Vector, Latent_State, Vintage_Adj)$$

In practice: the XGBoost / LightGBM Phase 2 model takes the full feature vector — origination PD, repayment ratio, payment efficiency, months since last payment, HMM posterior state probabilities, vintage adjustment factor — and outputs PD(t) as a calibrated probability.

Model Phases

Phase 1 — Static PD Model: Logistic regression, XGBoost, and LightGBM trained on origination features only (grade, sub_grade, fico, annual_inc, int_rate, purpose, term) with target = default. This establishes the origination PD baseline and benchmarks the dynamic model's incremental value.

Phase 2 — Latent State Integration: The HMM posterior state probabilities (P(Healthy), P(Watchlist), P(Stress), P(Critical)) are added as features. The dynamic model is trained on the full feature set including repayment progression metrics. This produces PD(t).

Phase 3 — Vintage Calibration: PD(t) is multiplied by the vintage adjustment factor to produce the final calibrated current PD.

Interactive: Illustrative PD Drift Demo — adjust repayment ratio to see how PD(t) evolves

Repayment Ratio

0.45

Months Since Last Payment

2

Grade (A=1, G=7)

B

Origination PD	13.39%
Current PD(t)	31.89%

State: Stress — Proactive Intervention

+18.50%

The Risk Drift Score is the central output of DRPR. It is the signed difference between the current PD and the origination PD:

Risk Drift Score		
Risk Drift = PD(t) - Origination PD		
Positive drift = borrower has deteriorated relative to underwriting expectations. Negative drift = borrower has improved (favorable early repayment). Magnitude encodes severity; sign encodes direction.		
Worked Example		
Borrower: Grade B, 36-month loan, issued 2015-Q2		
Origination PD (Grade B historical)		13.39%
Repayment Ratio at month 24 (expected: 0.65, actual: 0.31)	Below schedule	
Months since last payment	4 months	
HMM latent state	P(Critical) = 0.72	
Current PD(t)		38.22%
Risk Drift: Collections Escalation Triggered		+24.83%

Why Risk Drift Is Useful for Risk Officers

- **Grade-relative framing:** A PD of 25% means different things for a Grade A loan versus a Grade E loan. Risk Drift normalizes by the origination expectation, making cross-portfolio comparison meaningful.
- **Early warning signal:** Drift begins accumulating before a missed payment event. The repayment progression and latent state transitions provide leading rather than lagging indicators.
- **Portfolio heat maps:** Risk officers can visualize entire loan books as drift distributions, identifying segments, vintages, or origination cohorts experiencing systematic deterioration.
- **Provisioning support:** Loans with high positive drift require higher expected credit loss provisions. DRPR provides the quantitative basis for dynamic provisioning adjustments.
- **Regulatory evidence:** Under IFRS 9 and CECL frameworks, banks must demonstrate that their provisioning reflects current and forecast credit conditions. Risk Drift scores provide auditable, model-driven evidence of this assessment.

Any risk model deployed in an enterprise banking context must be explainable — to risk officers, to model validation teams, to regulators, and to internal audit. Black-box PD outputs without attribution are not acceptable in regulated financial services. DRPR uses SHAP (SHapley Additive exPlanations) throughout.

SHAP Application in DRPR

- **Feature-level attribution:** For each loan, SHAP values decompose PD(t) into the individual contribution of each input feature. A risk officer can see: "PD increased by 12% because months_since_last_payment increased by 3, and repayment_ratio fell below the grade-B expectation trajectory."
- **Drift attribution:** SHAP difference values between PD at t1 and PD at t2 identify which features drove the change in risk between two observation points.
- **Portfolio-level aggregation:** Mean absolute SHAP values across the portfolio identify the dominant risk drivers at any point in time — useful for macro risk reporting.
- **Watchlist explainability:** Every borrower placed on the watchlist receives a top-3 SHAP driver explanation, visible on the risk officer dashboard.

Regulatory Compliance Alignment

SHAP explainability supports compliance with: SR 11-7 (OCC/Federal Reserve model risk management guidance), IFRS 9 significant increase in credit risk (SICR) evidence requirements, CECL forward-looking loss estimation documentation, and GDPR/FCRA adverse action notice requirements (where applicable to consumer portfolios).

Model governance requirement: All models (Phase 1 static, Phase 2 dynamic, HMM latent state) must be registered in the model inventory, assigned model owners, and subject to annual independent validation. SHAP outputs form part of the model validation evidence package.

13 Intervention Engine

The Intervention Engine translates latent state and risk drift scores into tiered action recommendations. It operates as a rule-based system applied over the dynamic risk outputs.

Healthy	No Action. Loan performing within underwriting expectations. Drift ≤ 0. Continue standard monitoring cadence.	Drift ≤ 0%
Watchlist	Soft Reminder. Automated email / SMS reminder about upcoming payment. No human escalation. Log to watchlist queue. Review next cycle.	Drift 0–10%
Stress	Proactive Intervention. Relationship manager outreach. Hardship plan assessment. Potential restructuring options presented. SHAP report attached to case file.	Drift 10–25%
Critical	Collections Escalation. Transfer to collections queue. Legal review initiated. Collateral assessment (if secured). Provisioning flag raised to finance team.	Drift >25%

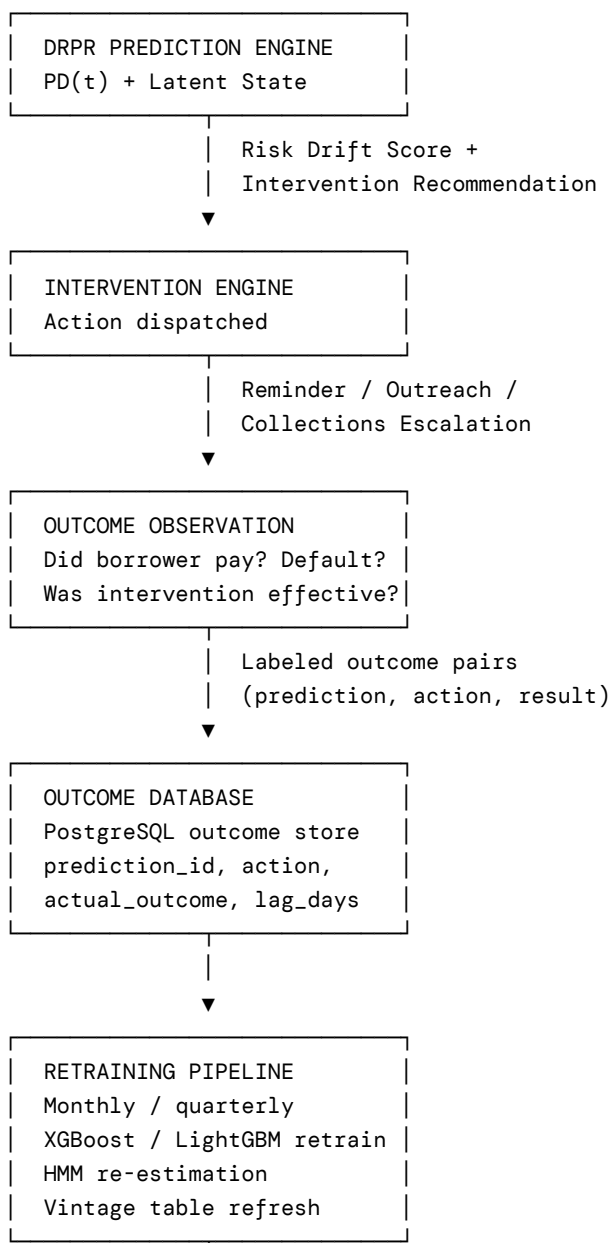
How Recommendations Are Generated

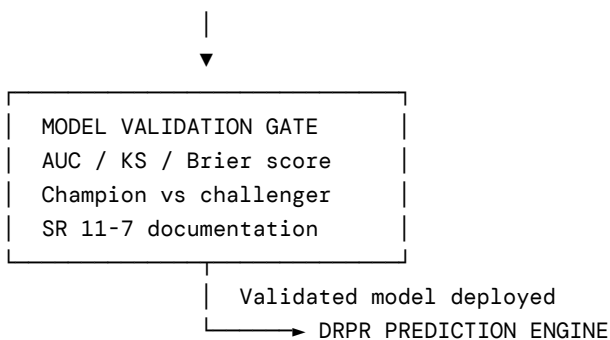
The intervention recommendation is a deterministic function of: (1) current latent state (HMM posterior argmax), (2) risk drift score, and (3) delta drift — the rate of change of drift over the past 3 observation periods. A borrower drifting rapidly from Watchlist toward Stress triggers an accelerated escalation path even before crossing the Stress threshold.

14 Feedback Loop

The system improves over time through a structured feedback loop that connects model predictions to real-world intervention outcomes and feeds results back into the training pipeline.

FEEDBACK LOOP ARCHITECTURE





Key Feedback Considerations

- **Intervention effect confounding:** Borrowers who received interventions are not a random sample. The feedback loop must account for selection bias — borrowers who were contacted may have different outcomes than similar borrowers who were not, independent of the intervention itself.
- **Data vintage lag:** Outcome data (whether an intervention prevented default) may lag 6–12 months. Retraining must account for incomplete follow-up periods in recent cohorts.
- **Champion/Challenger framework:** New model versions are deployed as challengers against the production champion, with A/B assignment and statistical monitoring before promotion.

15 Technology Stack

MVP STACK

Python 3.11	Pandas	NumPy	PostgreSQL
XGBoost	LightGBM	scikit-learn	SHAP
hmmlearn	Streamlit	FastAPI	SQLAlchemy

ENTERPRISE EXTENSION

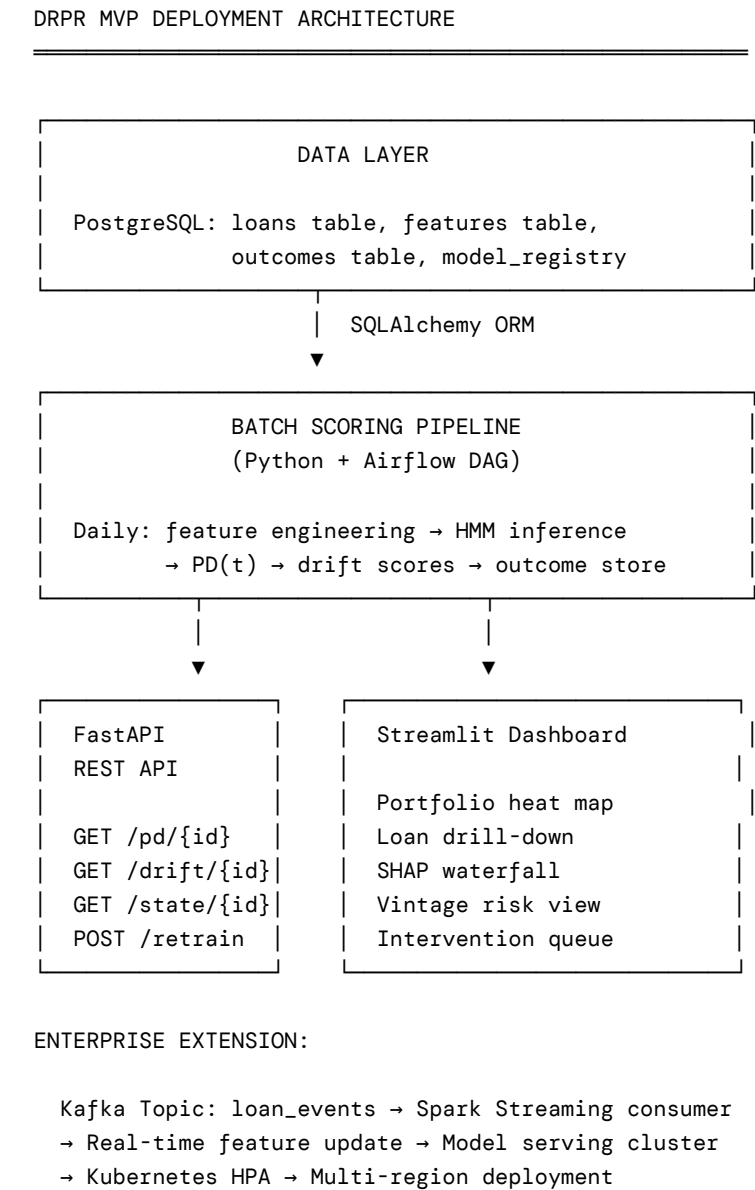
Apache Kafka	Spark Streaming	Docker	
Kubernetes	AWS SageMaker	Azure ML	MLflow
Airflow	Great Expectations	Grafana	
Prometheus			

Component Rationale

- **hmmlearn:** Python HMM library supporting Gaussian, MultinomialHMM, and GMMHMM models. Implements Baum-Welch training and Viterbi decoding. The GaussianHMM variant suits the continuous repayment signal emissions in DRPR.
- **XGBoost / LightGBM:** Industry-standard gradient boosting for tabular credit data. Both support calibrated probability outputs via Platt scaling and isotonic regression post-processors.
- **SHAP:** TreeExplainer for tree-based models provides exact Shapley values efficiently. KernelExplainer as fallback for the HMM layer.

- **FastAPI:** Async REST API layer serving PD(t) and drift score endpoints. Deployed as Docker container with Kubernetes horizontal pod autoscaling in enterprise version.
- **Streamlit:** Rapid risk officer dashboard deployment. Loan-level drill-down, portfolio heat maps, SHAP waterfall charts, vintage risk views.

16 **Deployment Architecture**



17 **Future Expansion**

FUTURE WORK ONLY. The following capabilities are not part of the MVP. They are described here to illustrate how the DRPR architecture could be extended once real-time data pipelines and external data sources become available. None of these should be conflated with the LendingClub-based MVP described above.

Real-time transaction streams (Open Banking / Account Aggregator): With access to checking account transaction streams (salary deposits, utility payments, discretionary spending), the repayment progression layer could be supplemented with direct income and expenditure signals. Sudden disappearance of salary deposits, spike in minimum balance periods, or increased overdraft frequency would become leading indicators of borrower stress — months before repayment behavior degrades.

Behavioral telemetry: Login frequency, mobile app engagement patterns, and digital banking behavioral signals correlate with financial stress in borrower populations. Integration of behavioral telemetry via a customer data platform (CDP) could enrich the HMM emission model significantly.

Alternative data sources: Bureau refresh data (credit score updates), property registry data for HELOC portfolios, employment verification via government databases, and social mobility indices could all improve vintage risk adjustment accuracy and latent state inference.

Real-time Kafka streaming architecture: In an enterprise deployment with continuous transaction data, DRPR would consume a Kafka topic of financial events, maintain a stateful feature store (Feast or Tecton), and serve PD(t) with sub-second latency via a model serving cluster. The HMM transition probabilities would be updated incrementally via online learning.

18 Risks and Limitations

- **Anchor dependency:** DRPR's risk drift framework is only as meaningful as the origination underwriting quality. If the origination PD anchor is systematically miscalibrated (e.g., grade inflation in a particular vintage), drift scores will be distorted. This is a known limitation of all anchor-relative monitoring systems.
- **Survivorship bias:** The LendingClub dataset contains only loans that were originated. Rejected applications are not visible, creating selection bias in any model trained on this data. The feature distributions of originated loans are systematically different from the full applicant population.
- **LendingClub platform specificity:** LendingClub's underwriting methodology, borrower base, and origination standards are specific to the US peer-to-peer consumer lending market. Direct transfer of model coefficients to a bank's mortgage, auto, or commercial lending portfolio would require significant recalibration.
- **No transaction-level data:** The MVP cannot observe actual cash flows, income continuity, or spending behavior. All borrower health inference is based on cumulative repayment aggregates, which are lagging

indicators relative to transaction-level data.

- **HMM stationarity assumption:** The Hidden Markov Model assumes stationary transition probabilities. In practice, transition dynamics change during economic recessions, interest rate cycles, and regulatory shifts. Periodic re-estimation is required.
- **Intervention causality:** Risk drift does not identify the cause of borrower distress — only its magnitude. Intervention design should not assume that contacting a borrower will necessarily address the underlying stress, which may be job loss, medical expense, or macroeconomic deterioration outside the borrower's control.
- **Regulatory risk (fair lending):** Any model that uses `addr_state`, `purpose`, or correlated features could introduce disparate impact on protected classes. Rigorous fair lending testing (adverse impact ratio analysis, ECOA/Regulation B compliance) is mandatory before enterprise deployment.

19 **Enterprise Banking Value Proposition**

LOSS PREVENTION	PROVISIONING	PORTFOLIO INSIGHT	REGULATORY
Earlier intervention on deteriorating credits	IFRS 9 / CECL alignment	Risk concentration visibility	Model governance ready
Moving from lagging to leading indicators can shift intervention timing 60–120 days earlier	Dynamic PD(t) provides the forward-looking ECL estimate required by current accounting standards	Aggregate drift scores identify vintage, segment, or origination channel deterioration before delinquency metrics	SHAP explainability, SR 11-7 documentation, champion/challenger framework all built in

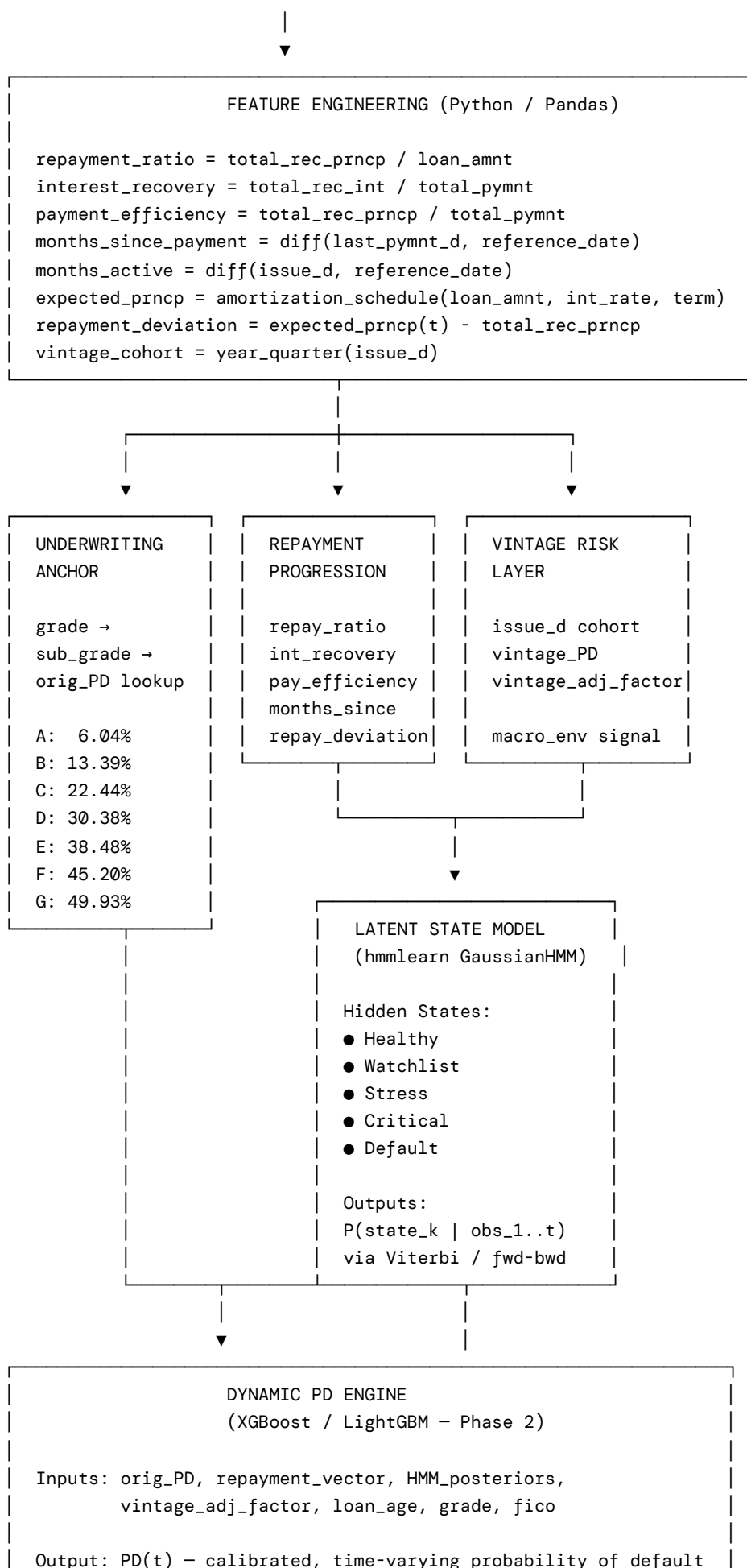
For tier-1 banks operating loan books in the tens of billions, even a 50 basis point improvement in charge-off rates through earlier intervention represents nine-figure P&L impact. The DRPR architecture is designed to be the infrastructure layer on which this improvement is built — incrementally, measurably, and with full model governance accountability.

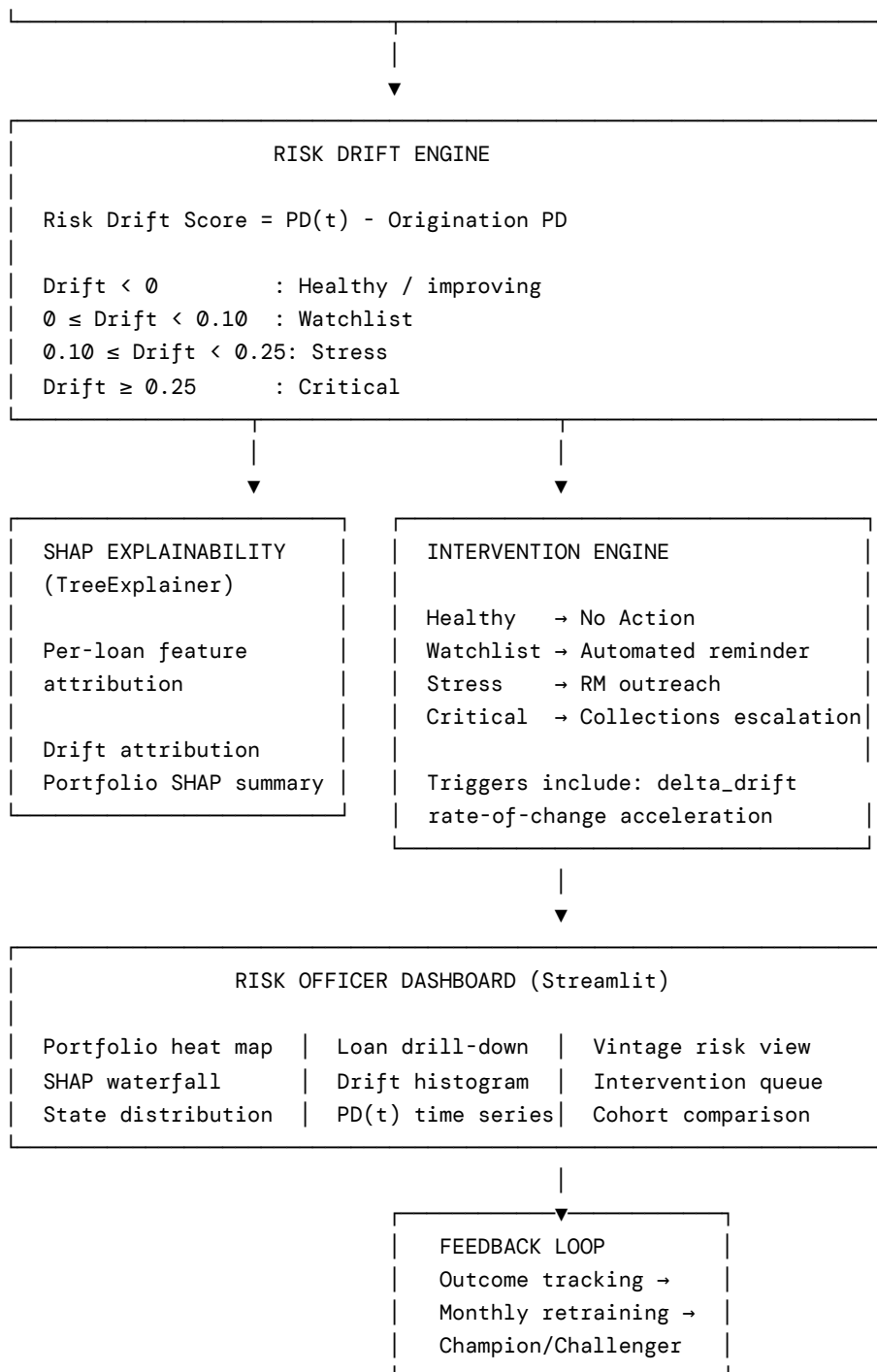
20 **Full ASCII Architecture Diagram**

```
DYNAMIC RISK PORTFOLIO REBALANCER – FULL SYSTEM ARCHITECTURE
=====

DATA INPUTS (MVP: LendingClub only)
=====
id loan_amnt term int_rate installment grade sub_grade emp_length
home_ownership annual_inc issue_d loan_status purpose addr_state
fico_range_low fico_range_high total_pymnt total_rec_prncp
```


total_rec_int last_pymnt_d





MVP STACK: Python · Pandas · NumPy · PostgreSQL · XGBoost · LightGBM
SHAP · hmmlearn · FastAPI · Streamlit · SQLAlchemy

ENTERPRISE: Kafka · Spark · Docker · Kubernetes · AWS/Azure · MLflow
